

Midterm Announcement

When and where: The midterm will be on Thursday, March 5th.

- A1 section, starting at 9:30 a.m.
 - **KCB 106:** Students whose first name starts with [A-L]
 - **SAR 103:** Students whose first name starts with [M-Z]
 - **CDS 614:** Students with exam accommodation
- A2 section, starting at 11:00 a.m.
 - **KCB 106:** Students whose first name starts with [A-I]
 - **HAR 222:** Students whose first name starts with [J-Z]
 - **CDS 263:** Students with exam accommodation

Crib sheet: The exam is *closed-book* and *closed-devices*. Leave unauthorized items, such as phones, calculators, headphones, earbuds, smartwatches, and any electronic device in your bag. You may bring **handwritten notes** on a 2-sided $8\frac{1}{2} \times 11$ or A4 sheet of paper. Preparing a crib sheet can be a useful study aid, so take time to select material for it.

Exam procedure: When you enter the room, we will ask you to leave your bag and coat at the podium and direct you to one of the seats. **We ask that you don't seat next to your friends / collaborators.** You will be allowed to take only your crib sheet, pens, pencils and erasers with you. We will provide scrap paper. **Please turn off your cell phone for the duration of the exam and leave it in your bag.** When the exams are distributed, please read the instructions on the front of the exam booklet. Do not open the booklet until instructed to do so. If you have questions at any point during the exam, raise your hand, and a staff member will assist you. If you are done with the exam before the exam period is over, please hand in your exam to an instructor, pick up your stuff, and **quietly leave the room.**

Material covered: You are responsible for all material covered in the lectures through the lecture on **03/03**¹, assigned reading for those lectures, homework 1–6, and discussions 1–6.

Practicing for the exam: Review lecture slides, homework problems, and discussion handouts. Ensure you have tried all solved exercises for the topics we covered in our second textbook (P). **At least one of the problems mentioned in the two previous sentences will appear on the exam.** The enclosed practice problems are meant to give you a general sense of what may be asked of you during the midterm. The actual exam will have *fewer problems* and of course, they will be different from the ones enclosed. **The goal of the practice exam below is to familiarize you with the midterm format, not to provide a comprehensive review of the topics covered on the midterm.**

¹The midterm will cover all topics you saw up to Lecture 11, inclusive.

- Skills:**
- You should be comfortable with the concepts from probability we studied and be able to use them to solve problems of the level you have seen in homework assignments.
 - Some questions will require you to be comfortable with material covered in the mathematical prerequisites. Examples include: combinatorics, arithmetic and geometric sums, derivatives and integrals.

Problem 1. (A game of cards) Consider the following experiment. Players numbered 1 through 6 each choose a different card from a 13-card deck consisting of just the spade suit. Then, players 1 and 2 compare their cards; the loser (holding the lower card) is eliminated. The winner of that comparison moves on and compares her card with player 3's card, and so on, until only one player (the one who chose the highest spade card initially) remains.

- (a) Let X be the random variable described by the number of times player 1 wins a comparison. For example, if player 1 has a card higher than player 2 and higher than player 3 but lower than player 4, then $X = 2$. Compute the PMF of X . In other words, Find $\Pr[X = i]$, for all possible values of i : $i \in [0, 5]$.
(Hint: it turns out it may be easier to find $\Pr[X \geq i]$ for all i first!)
- (b) Suppose now that player 1 wins \$5 for each comparison they won. Define an appropriate random variable W , in terms of X , for the total amount player 1 wins, and compute the CDF of W .

Problem 2. (Random hats) Suppose n people, each with a distinct hat, attend a party. Upon arriving at the party each person tosses their hat in a bin. Then, before leaving the party each person takes a random hat from the bin. In this problem, we will compute the PMF of the number of people that get their own hats back. This is an important discrete distribution with many real world applications.

- (a) Use the inclusion-exclusion principle to argue that the number of ways at least one person gets their own hat is

$$\binom{n}{1}(n-1)! - \binom{n}{2}(n-2)! \dots (-1)^{n-1} \binom{n}{n}(n-n)!.$$

- (b) Show that the expression in part (a) can be simplified to

$$n! \sum_{m=1}^n \frac{(-1)^{m-1}}{m!}.$$

- (c) Use part (b) to show that the number of ways that nobody gets their own hat is

$$n! \sum_{m=0}^n \frac{(-1)^m}{m!}.$$

We will call this number the n -th hat number.

- (d) For all $j \in \mathbb{N}$, let F_j denote the j -th hat number (that is, the number of ways nobody gets their own hat back at a party with j people). Argue that the number of ways that exactly k people get their own hat is $\binom{n}{k} F_{n-k}$.
- (e) In this part of the problem, we will use the approximation

$$\sum_{m=0}^n \frac{(-1)^m}{m!} \approx \frac{1}{e}$$

for sufficiently large n . Let X denote the number of people that get their own hat. Use part (c) and (d) to approximate the PMF of X for sufficiently large n .

Problem 3. (Continuous CDFs) Suppose you play a game where you throw a dart at a target 10cm in radius. Your dart is equally likely to land at any point inside the target. Let X be the distance of your dart in centimeters from the center of the target.

- (a) What is $\mathbb{P}[X = 3.0]$? Is this more or less than $\mathbb{P}[X = 8.0]$?
- (b) Give F , the CDF of X .
- (c) What is $\mathbb{P}[2.0 \leq X \leq 3.0]$? Is this more or less than $\mathbb{P}[7.0 \leq X \leq 8.0]$?
- (d) Define a function $S(X) = 100 - X^2$ that describes how many points you score for landing a dart X cm from the center. Give the CDF of $S(X)$.

Problem 4. (Allergic Reaction) Tiago has a tin with ten cookies and wants to eat 5 of them, one by one. Unfortunately for him, 2 of the cookies have peanuts in them! If Tiago eats both of the peanut cookies, he will have an allergic reaction and go to the hospital (which means he will stop eating cookies even if he hasn't eaten five yet).

- (a) Let P be the random variable which represents the number of peanut cookies Tiago eats. Find the PMF of P .
- (b) Let C be the random variable which represents the number of cookies (with or without peanuts) that Tiago eats. Find the PMF and CDF of C .

Problem 5. (Racing on a line) Alice and Bob are racing on a number line. Starting from a random position, their goal is to reach the origin, the position zero, before their opponent. Since Alice doesn't want to lose, Alice will run at twice Bob's speed whenever she is physically behind Bob (this means she is losing to Bob and running in the same direction). Otherwise, Alice runs at the same speed as Bob.

- (a) Bob is standing at position -3 , and Alice will be uniformly and randomly placed in the range $[-10, 10]$. What is the probability that the outcome will be a draw?
- (b) To make the race fairer, Alice and Bob are now being placed uniformly, randomly, and independently in the range $[0, 10]$. What is the probability that the outcome will be a draw?
- (c) Alice and Bob decide to move to a 2D plane to race. Alice draws a circular track on a 2D plane with a perimeter of 10 and marks one point on a track as a finish line. Then, Alice and Bob are placed uniformly, randomly, and independently on the track. Given that both Alice and Bob use the optimal path for this race, what is the probability that the outcome will be a draw?

Problem 6. (Tree diagrams) The New England Patriots and Seattle Seahawks have just entered overtime in a game of American football. The Patriots will possess the ball first. The rules of overtime in this sport are slightly complex (but they make for great tree diagrams!):

- When a team possesses the ball, they will either score a touchdown, score a field goal, or not score.
- When a team's possession is over, the other team gets the ball (unless the game is over).

- If a team scores a touchdown, then they win and the game is over.
 - If the Patriots don't score on their first possession, then whoever scores next wins and the game is over.
 - If the Patriots score a field goal on their first possession, then the Seahawks get the ball and must score.
 - If the Seahawks don't score, then they lose and the game is over.
 - If the Seahawks also score a field goal, then whoever scores next wins and the game is over.
 - If no one wins after three possessions, then time will run out, so the two teams tie and the game is over.
- (a) Create a complete tree diagram showing each team's possessions and the possible outcomes of those possessions.

Sports analysts have determined certain probabilities about how this overtime will proceed, but they only give you information for some of the events:

- Both teams score a field goal on their first possession: 0.15
 - Both teams don't score on their first possession: 0.09
 - The two teams tie: 0.072
 - The Seahawks never possess the ball: 0.2
 - The probabilities of each possession's outcomes are independent of other possessions.
- (b) Find the probability of each edge and each outcome of the tree diagram.
(Hint: start by determining the shape of the tree and which outcomes belong to each event.)

Problem 7. (Students problems) A student must choose from the subjects, art, geology, or psychology, as possible electives. Among available options, she is always equally likely to choose art or psychology and twice as likely to choose geology.

- (a) If she has to choose one elective, what are the respective probabilities that she chooses art, geology, and psychology?
- (b) If she has to choose two electives, what is the probability she chooses art and psychology?
- (c) For the two elective case, what is the probability she chose geology given that her other choice was art?

Problem 8. (Drink water) Among the students taking CS 237 this semester, 60% enjoy coffee, 70% enjoy tea, and 40% enjoy both. Prof. Tiago chooses a CS 237 student uniformly at random. What is the probability that the chosen student neither enjoys coffee nor tea?

Problem 9. (Random events) Let X be a number chosen uniformly at random from $[2, 10]$. Find the probabilities of the following events:

- (a) $X > 5$,
- (b) $5 < X < 7$,
- (c) $X^2 - 12X + 35 > 0$.

Problem 10. (What a great party) Suppose a group of 8 couples (16 people) are invited to a party. Each person is assigned a seat around a circular table uniformly and independently at random.

- (a) Count the number of distinct seating arrangements.
- (b) If Alice and Bob are both invited to the party, what is the probability that they are sitting next to each other?
- (c) Suppose Jack and Jill arrive at the party and see that Alice and Bob are sitting next to each other. Assuming Alice and Bob are in their assigned seats, what is the probability that Jack and Jill will be sitting next to each other?

Problem 11. (Mr. Smith and the kids) Mr. Smith has two children of different ages. (Assume boys and girls are equally likely.) Consider the following events:

$$\begin{aligned} A &= \text{"both children are boys,"} \\ B &= \text{"at least one child is a boy,"} \\ C &= \text{"older child is a boy."} \end{aligned}$$

- (a) Calculate probabilities for $\Pr(A \mid B)$ and $\Pr(A \mid C)$.
- (b) Let A and B be two disjoint events with $\Pr(A) > 0$ and $\Pr(B) > 0$. Are A and B independent?

Problem 12. (What a long line) Ten children (five boys and five girls) are standing in a line. All possible ways in which they might line up are equally likely. All children have different names. For each item below, compute the probability of the specified event.

- (a) Children appear in the line in alphabetical order by name.
- (b) All girls precede all boys.
- (c) Between any two girls there are no boys (that is, the girls stand together in an uninterrupted block).
- (d) Children alternate by gender in the line.
- (e) Neither the boys nor the girls stand together in an uninterrupted block.

Problem 13 (Gold and silver coins). You have two fair coins: silver and gold.

- (a) You toss each coin 6 times. Let S be the event that all tosses of the silver coin are heads. Let G be the event that all tosses of the gold coin are heads. Are S and G independent?

- (b) You roll a standard die. Let random variable D be the number (from 1 to 6) that you rolled. You toss each coin D times. Let S be the event that all tosses of the silver coin are heads. Let G be the event that all tosses of the gold coin are heads.
- (i) Draw a partial tree diagram for this random experiment. Include all outcomes for the die. For the coins, instead of including all outcomes, only mark whether S and G occurred. Include all branches where S and G occurred and omit the rest.
- (ii) Are S and G independent?
- (c) We perform the same random experiment as in part (b) and define D as before. Compute the probability that $D = 1$ given that we got exactly one heads from $2D$ coin tosses (D for the silver coin and D for the gold coin). **(Show your work and explain what laws of probability you are using.)**

Problem 14. (Go terriers!) The BU soccer team has two games scheduled this weekend. The team has probability 0.4 of not losing the first game, and it has probability 0.7 of not losing the second game, independently of the first. If the team does not lose a particular game, it is equally likely to win or tie. The team receives 2 points for a win, 1 point for a draw, and 0 points for a loss. Let X be the number of points that the team wins this weekend. Find the PMF and CDF of X .

Problem 15. (PDF or not a PDF? That is the question) For each of the following piece-wise functions, determine whether or not it is a valid PDF for a continuous random variable. If it isn't, please explain why.

(a)

$$f_X(x) = \begin{cases} 0 & x < 0 \\ \frac{1}{3}x & 0 \leq x \leq 3 \\ 1 & x > 3 \end{cases}$$

(b)

$$f_X(x) = \begin{cases} \frac{1}{10} & 0 \leq x < 6 \\ \frac{1}{5} & 6 \leq x \leq 8 \end{cases}$$

(c)

$$f_X(x) = \begin{cases} -\frac{2}{3}x + \frac{4}{3} & 0 \leq x < 3 \\ 0 & \text{otherwise} \end{cases}$$

(d)

$$f_X(x) = \begin{cases} \frac{1}{6}x & 0 \leq x < 3 \\ 250 & 3 \leq x \leq 3.001 \\ 0 & \text{otherwise} \end{cases}$$

(e)

$$f_X(x) = \begin{cases} 0.3 & x = 3 \\ 0.7 & x = 5 \\ 0 & \text{otherwise} \end{cases}$$

Problem 16. (CDFs and PDFs) A continuous random variable M has the following PDF:

$$f_M(x) = \begin{cases} x^2 - 2x + 1 & \text{if } 1 \leq x \leq 2 \\ \frac{1}{3} & \text{if } 5 \leq x \leq 7 \\ 0 & \text{otherwise} \end{cases}$$

Find:

- (a) the CDF of M ,
- (b) $\Pr\left(\frac{1}{2} \leq M \leq 6\right)$.

Problem 17. (Two decks of cards) A deck of cards consists of 6 red and 5 black cards. A second deck of cards consists of 9 red cards. A deck is selected uniformly at random and a card is drawn uniformly at random from the deck and found to be red. What is the probability that the card was drawn from the first deck?

Problem 18. (High interest rates) Defaulting on a loan means failing to pay back the loan. In ProbabilityVille, 5% of the residents will default on their loan. The Central Bank of ProbabilityVille has developed a test to predict whether a person will default. The test is not perfect though:

- If the person will default, there is a 6% chance that the test will predict that the person will not default.
- If the person will not default, there is a 2% chance that the test will predict that the person will default.

A person is chosen uniformly at random from the residents of ProbabilityVille. Given that the test predicts that the person will default, what is the probability that the person will truly default?

Problem 19. (Playing with cards) We have a deck of cards (52 in total). Let's play some card drawing games.

- (a) What is the probability of drawing 5 hearts with replacement?
- (b) What is the probability of drawing 5 spades without replacement?
- (c) Given that we already have 2 Kings in hand, what is the probability of drawing another three cards with the same rank from the remaining 50 cards without replacement?

Problem 20 (Proof problem). Fill in the blanks to correctly complete the proof of the following statement:

Let Ω denote a set of all events, and let $\mathbb{P}$ be a probability function defined over it, such that the probability of any event $A \subset \Omega$ is given by $\mathbb{P}[A]$.

Lemma. *The probability of the empty event is $\mathbb{P}[\emptyset] = 0$.*

Proof. If $\{A_1, A_2, \dots, A_n\}$ is a sequence of events such that $A_i \cap A_j = \emptyset$ for all $i, j \leq n$, then $\mathbb{P}[A_1 \cup A_2 \cup \dots \cup A_n] = \text{}$.

The previous statement implies that $\mathbb{P}[\Omega \cup \emptyset] = \boxed{}$, since Ω and $\emptyset$ are $\boxed{}$ sets by definition, i.e. $\Omega \cap \emptyset = \boxed{}$.

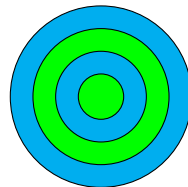
Since $\Omega \cup \emptyset = \boxed{}$, we have $\mathbb{P}[\Omega \cup \emptyset] = \boxed{} = 1$. Therefore, $\mathbb{P}[\Omega \cup \emptyset] = \boxed{} = \mathbb{P}[\Omega] = 1$, so $\mathbb{P}[\emptyset] = 0$.

□

Problem 21. (Race Preparation) Alice is preparing for her upcoming 400 meters race. Since Alice doesn't know how fast she should run, she decides to choose her speed uniformly and randomly in range $[5, 10]$ meters per second and run at that constant speed until she reaches the end of the race.

- What is the CDF of T , which is the amount of time (in seconds) it takes for Alice to run 400 meters?
- Since she doesn't have much time to prepare for the race, Alice decides to run twice a day and record the time of the slower run in her record. What is the CDF of S , which is the time of the slower run?
- Recording the slower run makes Alice hesitate to run fast. Thus, she decides to record the time of the faster run instead. What is the CDF of Q , which is the time of the faster run?

Problem 22 (Dartmaster). We toss a dart at a target that consists of 4 concentric circles of radius 1, 2, 3, and 4 inches, respectively. The areas between the circles are colored as shown below. Find the probability that the dart lands within 3 inches of the center of the target, given that it lands in a blue region.



Problem 23 (Cheat Code). Alice wants to impress her friends with her gaming skills so she tries to activate a cheat code. The only problem is that she couldn't remember the cheat code! She only remembers some information about it. Help her recover the cheat code using the following information:

- The code only consists of U and D .
- The code contains at least 1 U and 1 D .
- There are no 3 consecutive D .
- The probability that DD is immediately followed by a U is $\frac{2}{3}$
- The probability that U is immediately followed by a D is $\frac{4}{5}$
- The probability that U is immediately preceded by a D is $\frac{3}{5}$

7. For all suffixes of the cheat code, the probability that we get a D from uniformly and randomly sampling a character from the suffix is either 0, 1, in the range $[\frac{1}{8}, \frac{2}{5}]$, or in range $[\frac{7}{12}, \frac{7}{9}]$.
- (a) Use information 2, 5 and 6 to find the number of U in the cheat code.
 - (b) Use information 3 and 4 to find the number of DD in the cheat code.
 - (c) Use all information you have to recover the cheat code.
(Hint: Starts from the back of the cheat code.)